

Consider the data points in 2-D space of the following Figure 1.

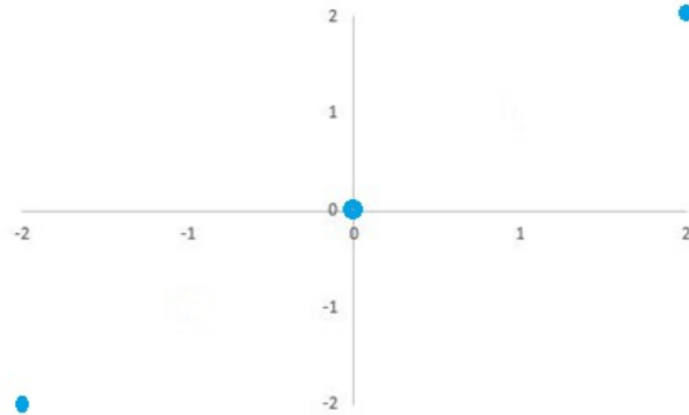


Figure 1: Dataset of the Points A(0,0), B(2,2), C(-2,-2).

Apply principal components analysis (PCA) to determine the first principal component for the

dataset in Figure 1. Note that, you have to determine the eigenvectors $\begin{pmatrix} x_i \\ x_{i+1} \\ \vdots \\ x_n \end{pmatrix}$ and afterward, each

of the eigenvectors must be scaled to unit length eigenvector $\begin{pmatrix} x'_i \\ x'_{i+1} \\ \vdots \\ x'_n \end{pmatrix}$ using the equation $x'_i =$

$\frac{x_i}{\|x\|_2}$, Where, $\|x\|_2$ is the l_2 norm of vector $x = [x_i, x_{i+1}, \dots, x_n]$.

x	y
0	0
2	2
-2	-2
Σ	0

$x_i - \bar{x}$	$y_i - \bar{y}$
0	0
2	2
-2	-2
Σ	0

$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
0	0	0
4	4	4
4	4	4
8	8	8

①

$$\text{covar} = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$\text{Covar matrix } A = \begin{pmatrix} x & y \\ 4 & 4 \\ 4 & 4 \end{pmatrix}$$

$$\textcircled{2} \quad \lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$\textcircled{3} \quad A - \lambda I = \begin{pmatrix} 4-\lambda & 4 \\ 4 & 4-\lambda \end{pmatrix}$$

$$\begin{aligned} \textcircled{4} \quad |A - \lambda I| &= 16 - 4\lambda - 4\lambda + \lambda^2 - 16 \\ &= \lambda^2 - 8\lambda \end{aligned}$$

$$\therefore \lambda^2 - 8\lambda = 0$$

$$\lambda_1 = 8$$

$$\lambda_2 = 0$$

$$\textcircled{5} \quad \lambda_1 = 8 :$$

$$A - \lambda I = \begin{pmatrix} 4-8 & 4 \\ 4 & 4-8 \end{pmatrix}$$

$$B = \begin{pmatrix} -4 & 4 \\ 4 & -4 \end{pmatrix}$$

$$\textcircled{6} \quad B \cdot x = 0$$

$$B \quad R_2 \rightarrow R_1 + R_2$$

$$B = \begin{pmatrix} -4 & 4 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-4x_1 + 4x_2 = 0$$

$$x_1 = \frac{4x_2}{4} = x_2$$

$$x_1 = 1, \quad x_2 = 1$$

$$\vec{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\lambda_2 = 0,$$

$$A - \lambda I = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

$$B = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

$$B \rightarrow R_2 - R_1$$

$$\begin{pmatrix} 4 & 4 \\ 0 & 0 \end{pmatrix} \times \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$4x_1 + 4x_2 = 0$$

$$x_1 = -x_2$$

$$\bar{x} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$A \bar{x} = \lambda \bar{x}$$

$$\frac{-1}{\sqrt{1+1} + i\sqrt{1+1}}$$

$$F = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 \\ 2 & 2 \\ -2 & -2 \end{pmatrix}$$

$$FD = F^T D^T$$

$$= \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \times \begin{pmatrix} 0 & 2 & -2 \\ 0 & 2 & -2 \end{pmatrix}$$

2x2

2x3

2x

$$= \begin{pmatrix} 0 & \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} & -\frac{2}{\sqrt{2}} - \frac{2}{\sqrt{2}} \\ 0 & -\frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} & \frac{2}{\sqrt{2}} - \frac{2}{\sqrt{2}} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2.82 & -2.82 \\ 0 & 0 & 0 \end{pmatrix}$$

$$PD = (FD)^T$$

$$= \begin{pmatrix} 0 & 0 \\ 2.8 & 0 \\ -2.8 & 0 \end{pmatrix}$$

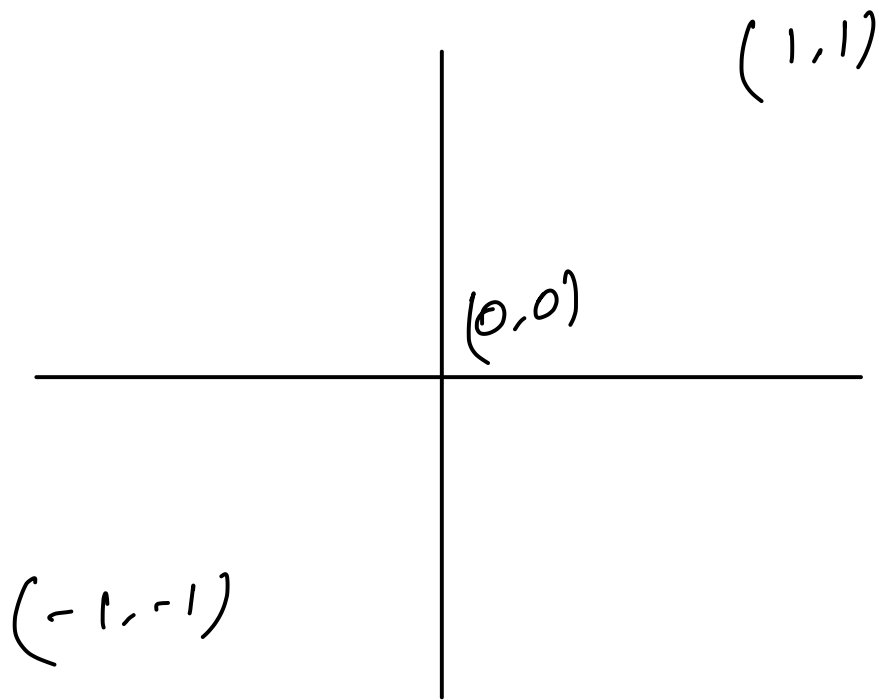
$$\lambda_1 = 8$$

$$\lambda_1 = \frac{8}{8} = 100\%$$

$$\lambda_2 = 0$$

$$\lambda_2 = \frac{0}{8} = 0\%$$

$$\text{Total} = 8$$



x	y	$x_i - \bar{x}$	$y_i - \bar{y}$
0	0	0	0
1	1	1	1
-1	-1	-1	-1
0	0		

Ang

$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
0	0	0
1	1	1
1	1	1
2	2	2

(i)

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

(ii)

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{pmatrix} = \begin{pmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{pmatrix}$$

$$\begin{aligned} (iii) |A - \lambda I| &= (1-\lambda)(1-\lambda) - 1 \\ &= \lambda^2 - 2\lambda = 0 \end{aligned}$$

$$\lambda = 2$$

$$\lambda = 0$$

$$\text{If } \lambda = 2$$

$$A - \lambda I = \begin{pmatrix} 1-2 & 1 \\ 1 & 1-2 \end{pmatrix}$$

$$B = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$B \rightarrow R_2 \rightarrow R_2 + R_1$$

$$B \rightarrow \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \times \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-x_1 + x_2 = 0$$

$$n_1 = n_2$$

$$\therefore \text{if } n_2 = 1 \quad n_1 = 1$$

$$n_2 = 2 \quad n_1 = 2$$

$$\bar{X} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\Rightarrow 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\bar{X} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{\text{Normalized}}$$

$$\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

Test

$$A \bar{X} = \lambda \bar{X}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2×2

$$= \begin{pmatrix} 1+1 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

✓

Now if $\lambda = 0$

$$A - \lambda I - B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$B \rightarrow R_2 - R_1$$

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} * \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

$$x_2 = 1$$

$$\bar{x} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Normalized

$$\frac{\begin{pmatrix} -1 \\ 1 \end{pmatrix}}{\sqrt{1^2 + (-1)^2}}$$

$$A(\bar{x}) = \lambda \bar{x}$$

$$= \begin{pmatrix} -1 \\ 1 \end{pmatrix} * \frac{1}{\sqrt{2}}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$F = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$FD = F^T \cdot D^T$$

$$= \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} *$$

2x2/

2x3

$$\begin{pmatrix} 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix}$$

$$= 0 \quad \frac{\sqrt{2}}{2} * \frac{\sqrt{2}}{2}$$

$$= \begin{pmatrix} 0 & 1.4 & -1.4 \\ 0 & 0 & 0 \end{pmatrix}$$

$$PD = (FD)^T$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 1.4 & 0 & 0 \\ -1.4 & 0 & 0 \end{pmatrix}$$

New Dataset

X	Y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
Avg = 1.775	2.05

mean subtraction

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
0.725	0.35	0.253
-1.275	-1.35	1.721
0.425	0.85	0.361
0.125	0.15	0.01875
		2.35

$(x_i - \bar{x})$	$(y_i - \bar{y})$
0.525	0.122
1.625	1.82
0.18	0.72
0.015	0.022
2.345	-0.956

2.684

$$\textcircled{1} \text{Cor}(A) = \begin{matrix} x & y \\ \begin{pmatrix} 0.781 & 0.783 \\ 0.783 & 0.89 \end{pmatrix} \end{matrix}$$

$$\textcircled{2} A - \lambda I = \begin{pmatrix} 0.78 - \lambda & 0.78 \\ 0.78 & 0.89 - \lambda \end{pmatrix}$$

$$\textcircled{2} \quad |A - \lambda I| = -0.241 + 0.31\lambda - \lambda 0.78 + \lambda^2$$

$$= \lambda^2 - 0.47\lambda - 0.241 = 0$$

$$\Rightarrow \lambda_1 = 0.77$$

$$\lambda_2 = -0.3$$

$$\textcircled{4} \quad \text{when } \lambda_1 = 0.77$$

$$A - \lambda I = \begin{pmatrix} 0.78 - 0.77 & 0.78 \\ 0.78 & -0.31 - 0.77 \end{pmatrix}$$

$$B = \begin{pmatrix} 0.01 & 0.78 \\ 0.78 & -1.08 \end{pmatrix}$$

⑤

$$0.78 - \lambda 0.01 = 0$$

$$\Rightarrow R_2 - 78 R_1$$

$$= \begin{pmatrix} 0.01 & 0.78 \\ 0 & -61.92 \end{pmatrix}$$

$$-61.92 + \lambda 0.78 = 0$$

$$\Rightarrow \lambda = \frac{61.92}{0.78} = 79.38$$

$$\Rightarrow R_2 + 79.38 R_1$$

$$\begin{pmatrix} 0.01 & 0.78 \\ 0 & 0 \end{pmatrix} \times \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$0.01 x_1 + 0.78 x_2 = 0$$

$$\Rightarrow x_1 = \frac{-0.78 x_2}{0.01}$$

$$\text{if } x_2 = 1$$

$$x_1 = -78$$

$$\vec{v} = \begin{pmatrix} -78 \\ 1 \end{pmatrix}$$

maheer Helal

		$X_i - \bar{X}$	$Y_i - \bar{Y}$
	X		
	-1	-1	-0.67
	0	0	0.33
	1	1	0.33
mean	0		
			0.67

$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	0.448	0.67
0	0.108	0
1	0.108	0.33
2	0.664	1

$$A = \begin{pmatrix} x & x & y \\ 1 & & 0.5 \\ y & 0.5 & 0.332 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & 0.5 \\ 0.5 & 0.33 - \lambda \end{pmatrix}$$

$$|A - \lambda I| = 0.33 - 0.33\lambda - \lambda + \lambda^2 - 0.25$$

$$= \lambda^2 - 0.67\lambda - 0.08 = 0$$

$$\lambda = 0.51$$

$$\lambda_2 = 0.15$$

$$\lambda_1 = 0.51 :$$

$$A - \lambda I = \begin{pmatrix} 1 - 0.51 & 0.5 \\ 0.5 & 0.33 - 0.51 \end{pmatrix}$$

$B =$

$$= \begin{pmatrix} 0.49 & 0.5 \\ 0.5 & -0.18 \end{pmatrix}$$

$B \Rightarrow$

1

1

0

0

- 1

- 1

0

0

